

Clock Page Replacement Algorithm

1 Core Idea

Definition

The **clock page replacement algorithm** is an efficient implementation of second chance. It keeps page frames in a circular list and uses a moving hand to find a victim.

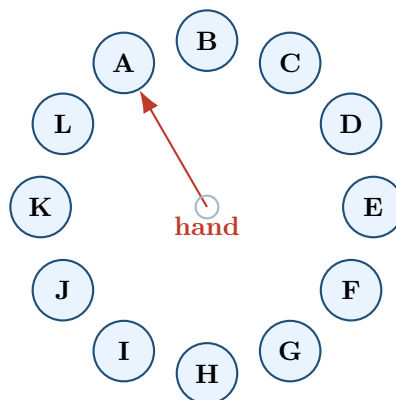
Second chance moves pages around in a list. Clock avoids that overhead by leaving pages in place and moving only a pointer, called the **clock hand**.

Clock Rule

$R = 0$: evict the page under the hand
 $R = 1$: clear R and advance the hand

2 Clock Structure

All page frames are arranged in a circular list.



Meaning of the Hand

The hand points to the oldest candidate page. On a page fault, the OS starts checking from that position.

3 Algorithm Steps

When a page fault occurs:

1. Inspect the page currently under the clock hand.
2. If $R = 0$, evict that page.
3. Insert the incoming page into the same frame.

4. Advance the hand to the next frame.
5. If $R = 1$, clear R , advance the hand, and continue scanning.

4 Decision Table

R bit	Action
0	The page has not been referenced recently. Evict it.
1	The page was referenced recently. Clear R , give it a second chance, and move the hand forward.

5 Worked Example

Problem

The hand points to page A . The pages have these referenced bits:

Page	A	B	C	D	E
R	1	1	0	1	0

Which page is evicted?

Solution

Checked page	R	Action
A	1	Clear R , advance hand.
B	1	Clear R , advance hand.
C	0	Evict C .

Answer

Page C is evicted because it is the first page found with $R = 0$.

6 Why Clock Is Better Than Moving Lists

Second chance and clock make almost the same replacement decision. The difference is implementation cost.

Algorithm	Structure	Cost issue
Second chance	Linear FIFO list.	Pages may be moved to the end of the list.
Clock	Circular list with a hand.	Pages stay in place; only the hand moves.

Efficiency Point

Clock avoids repeatedly rearranging the page list, so it is a cleaner implementation of second chance.

7 If All Pages Have $R = 1$

Clock still terminates.

1. The hand scans pages one by one.
2. Every page with $R = 1$ has R cleared.
3. After one full rotation, pages previously seen now have $R = 0$.
4. The next suitable page is evicted.

Degeneration

If every page was recently referenced, clock eventually behaves like FIFO because all pages receive a second chance.

8 Final Summary

One-Box Summary

Clock is second chance implemented with a circular list.
The hand scans pages and checks the R bit.
If $R = 0$, evict; if $R = 1$, clear it and advance.
It avoids moving pages around like second chance.